Errors, p-values, and Significance Transcript

Hello everyone, welcome to lesson 2 video 2. In this we will explore errors, p-values and level of significance in hypothesis testing. These are key ideas that determine how we interpret results of a test. The main topics that we'll be including in this video are type 1 error, type 2 error and then various different aspects of it.

So far we have built the framework for hypothesis testing and learned how to apply it right. Today we are zooming into one of the most critical part of the process, the final decision making. This session is all about the art of statistical judgement.

We will take a deeper look at the errors we can make, clarify the meaning of p-value, we define what it truly means for a result to be statistically significant. And this is where all the pieces come all together. So let's get started.

As we discussed hypothesis testing is like courtroom trial right. If you remember that's what we went through in the last few videos. We assess evidence from a sample to reach a verdict on a claim about a population.

We decide to either reject or fail to reject the null hypothesis. This is what we eventually do. We either say that we are rejecting null hypothesis or we fail to reject the null hypothesis right.

But it's very crucial to remember that we are working within incomplete information. A sample is just a snapshot but not the whole picture right. It's just a piece of information from the entire population.

Because of this our verdict is never certain. There is always a risk that our sample is misleading and then in that case our conclusion would always be wrong right. Understanding and managing this risk is what separates casual behaviour or casual observation from rigorous scientific decision making.

So we would be doing a lot of the studies around these to understand that a sample may be misleading in certain situations and that is where we would have to be very careful. So there are these different two types of errors that we already have discussed. The very first error type one error this is a false alarm right.

Type one error is called is a false alarm because this is when we reject the null hypothesis even though it was actually true. Think of a fire alarm right for ringing when there is no fire. So you have a fire alarm when the fire is actually not there or a medical screening flagging a healthy person for a disease.

So you are saying that you have went out for a test and the test says you are positive for a particular disease when actually you are not positive right. So this basically is the type one error. This is a false alarm.

So you can also quantify or you can also write type one error as a false alarm and the simplest possible explanation to you which can help you relate to it is you go out for testing for various different types of tests and somehow the test shows you that you are having a particular disease but you may not be having that disease. So test could also be wrong right. That is a false alarm.

Type two error is misdetection. Now let's say you went out for the test and test results gave you that you are not affected though you are actually having the disease. That's also a mistake right.

So there are these two types of mistakes that we have one false alarm second the misdetection. So these are the type one and type two error. So the analogy is what I gave you and I think you have understood well right.

So what actually happens when you have a type one error or type one error is very significantly related to the level of significance your line in the stand. So setting the standard of proof what do you mean by that is we always assume or we take a level of significance by assuming about how much amount of confidence do we want in our entire testing mechanism we are having. So that alpha values level of significance is chosen as either 0.5 0.05 or 0.1 or 0.001 either 1 percent or 5 percent or 10 percent.

Generally these are the values that we take. So we restrict our alpha values majorly in 90 95 percent of situations to 5 percent which means that I am willing to accept up to 5 percent of chances of type one error which is if you remember the raising of false alarm right. So you are saying that I can only give leverages to false alarms up to 5 percent OK.

So that's what your alpha value is and type one error is as all related to your alpha value. That's what I just wanted to help you understand. Now comes the P value.

This is also very interesting and very important part of entire hypothesis testing framework. So alpha is our pretest standard for the for for proof right. The P value is the actual strength of evidence we find.

So we first actually frame our standards. We say that this is our standard. If our standard are missed we will look into it.

Now P value is after you perform your test you are trying to find out whether we have missed our standard or not. OK that's what your P values are. While alpha is a preset standard the P value is what we calculate from our data.

OK it measures the strength of our actual evidence right. Let's revisit the core idea. The P value answers the question if the null hypothesis is true what's the probability of getting a result like mean right or even more extreme just by random luck right.

So what exactly that mean. Think of this like testing a coin and you assume that the coin is fair. If you flip it 10 times you get 6 heads.

That's not a weird output right. It could be weird. But for me it is not weird because out of 10 you are getting 6 heads which is very close to 50 50 right.

And that's why I'm saying it's not very weird. The P value would be very high indicating that your evidence is weak. But if you get 10 heads in a row you're almost certain that or you would be concluding that the coin is unbiased sorry the coin is biassed.

It's not a fair coin. It is biassed towards the heads right because it is resulting in heads out of the 10 trials if you get heads 10 times then obviously your coin is an unfair coin. That statement does not hold true right.

So in that case you would always be looking for these types of situations right. So when you are testing a coin assuming that the coin is fair and you flip that 10 times you get heads 10 number of times then in that case you are thinking that your assumption about the coin is fair is under the test right which is where you would start doubting your assumption. The P value would be tiny in that case indeed indicating that there is a very strong evidence that the coin itself is unfair.

So that's what it means. This is all the analogy I have written down here which is what I was trying to explain you right. The crucial point is P value is not the probability that the null hypothesis is true right.

So P value is actually the level of significance at which we can reject null hypothesis. It's not the probability that the null hypothesis is true. It's the probability at which we reject null hypothesis.

Now P value versus alpha this is also was covered in one of the videos if you remember it's just that I wanted all of us to go through it in much more details. Now this also is the climax of the test. The decision itself is simple direct comparison.

We take strength of evidence the P value we calculate from our data and we compare it to our predetermined standard that we have set right. So remember alpha is a standard that we have set. We say that the possibility of detecting false alarm is 5 percent.

OK. Then if we get P value less than 5 percent we say that we have already given very less possibilities of detecting false alarm and if we get a value in that range up to 5 percent it means that that is not by chance. It means there has to be some core evidence or some significance in the values that we have got.

Right. If after doing the testing the P value comes out to be less than 5 percent in that case you say that the evidence is strong and hence it is statistically significant we reject H naught. If P value is greater than alpha it means that there is no strong evidence not statistically significant and we fail to reject H naught.

So that's what happens in various different scenarios. This is just a simulation to help you understand when I change the level of significance here you will see that based on this is your standard the level of significance if you see once you change your standard okay this is this is what right once you change your standard you will see that if you are if your P value in this scenario right now this is your P value this is your standard because your P value is greater than your standard you are good I mean you do not have level of significant decision making or a significant outcome or significant evidence. Hence you are saying that your alternate sorry your null hypothesis is true.

But as soon as you change it if you see here as soon as your standard okay in this case if you if you see here as soon as your standard is greater than okay if you see if your standard is greater than the P value in those cases you are saying that the null hypothesis is not true and the alternate hypothesis is true okay and you can always change the observed mean and you can look for various different scenarios this is what I also covered in one of the sessions but herein I just wanted to help you understand that you have the standard okay let's say for example this is the standard now this is the standard for you okay if the P value is lesser than standard this is the darker region if the P value is lesser than the standard okay in that case what it means it means that the evidence that you have got is significant okay and in that case it means that whatever your values are you are moving towards alternate hypothesis okay so this is all what I wanted to help you understand I hope this was easier and intuitive for all of you to understand and you can always play around with this on the statistics so this is this is a simple summary of what we have gone through we have learned about P value we have learned about the two different types of errors and so on in summary understanding others P values helps us avoid false conclusion and quantify our results these tools makes hypothesis testing both rigorous and interpretable in the next video we'll apply this to one sample tests

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